

题型

1. 填空 (13道, 26分)

2. 计算 (6道, 48分)

3. 综合 (计算 + 证明) (26分)

完全为了期末速通

只用来解题.

不讲原理. 和定理

欧拉定理 $a^{\varphi(m)} \equiv 1 \pmod{m}$

费马小定理 $a^p \equiv a \pmod{m}$

最大公因数 — 广义欧几里得 (辗转相除)

$$r_{-2} = a \quad r_{-1} = b \Rightarrow (a, b) = r_n$$

$$r_{-2} = q_0 \cdot r_{-1} + r_0 \quad a = r_0 \pmod{b}$$

$$r_{-1} = q_1 \cdot r_0 + r_1 \quad b = r_1 \pmod{r_0}$$

$$r_0 = q_2 \cdot r_1 + r_2 \quad r_0 = r_2 \pmod{r_1}$$

⋮

最大公因数.

$$r_{n-2} = q_n \cdot r_{n-1} + r_n \quad r_{n-2} = r_n \pmod{r_{n-1}}$$

$$r_{n-1} = q_{n+1} \cdot r_n + \underline{r_{n+1}} = 0 \quad \text{直到余数为0}$$

$$\boxed{a, b} : \dots r_0$$

$$(1613, 3589)$$

$$(2947, 3772)$$

$$3589 = 2 \times 1613 + 363$$

$$3772 = 1 \times 2947 + 825$$

$$1613 = 4 \times 363 + 161$$

$$2947 = 3 \times 825 + 472$$

$$363 = 2 \times 161 + 41$$

$$825 = 1 \times 472 + 353$$

$$161 = 3 \times 41 + 38$$

$$472 = 1 \times 353 + 119$$

$$41 = 1 \times 38 + 3$$

$$353 = 2 \times 119 + 115$$

$$38 = 12 \times 3 + 2$$

$$119 = 1 \times 115 + 4$$

$$3 = 1 \times 2 + \textcircled{1}$$

$$115 = 28 \times 4 + 3$$

$$2 = 2 \times 1 + 0$$

$$4 = 1 \times 3 + \textcircled{1}$$

↑ 系数
↑ 除数
↑ 余数

$$3 = 3 \times 1 + 0$$

贝祖等式——逆向辗转相除，求系数

$$s \cdot a + t \cdot b = (a, b)$$

eg. $(169, 121)$

$$169 = 121 \times 1 + 48$$

$$121 = 48 \times 2 + 25$$

$$48 = 25 \times 1 + 23$$

$$25 = 23 \times 1 + 2$$

$$23 = 2 \times 11 + 1$$

$$2 = 1 \times 2 + 0$$

$$1 = 23 + (-11) \times 2$$

$$= 48 + (-1) \times 25 + (-11) \times [25 + (-1) \times (48 + (-1) \times 25)]$$

$$= 169 - 121 - 121$$

从上至下，分解余数，用 a, b 表示，层层代入

$$169 = 121 \times 1 + 48$$

$$121 = 48 \times 2 + 25$$

$$48 = 25 \times 1 + 23$$

$$25 = 23 \times 1 + 2$$

$$23 = 2 \times 11 + 1$$

$$2 = 1 \times 2 + 0$$

$$48 = 169 - 121$$

$$25 = 121 - 2 \times 169 + 2 \times 121 = 3 \times 121 - 2 \times 169$$

$$23 = 169 - 121 - 3 \times 121 + 2 \times 169 = 3 \times 169 - 4 \times 121$$

$$2 = 3 \times 121 - 2 \times 169 - 3 \times 169 + 4 \times 121 = 7 \times 121 - 5 \times 169$$

$$1 = 3 \times 169 - 4 \times 121 - 7 \times 121 + 55 \times 169$$

$$1 = 58 \times 169 - 81 \times 121$$

$$\Rightarrow -81 \times 121 + 58 \times 169 = (121, 169) = 1$$

$$312^2 \equiv 629 \pmod{667}$$

一次同余式求解

$$8 \equiv 1 \pmod{7}$$

$$ax \equiv b \pmod{m}$$

1. 先判断有无解.

- $(a, m) | b$ 有解
- 若 $(a, m) = 1$, 有唯一解.
- $(a, m) \nmid b$ 无解.

2. 求 $ax \equiv 1 \pmod{m}$

据上式求解出 $x_0' \equiv ? \pmod{m}$

3. 乘回系数 b

$$ax \equiv b \pmod{m}$$

4. 若 $(a, m) = 1$, 即只有 1 解时

$$x = bx_0 \pmod{m}$$

否则代入公式求解

$$x = \frac{b}{(a, m)} \cdot x_0 + \frac{m}{(a, m)} \cdot t$$

$$t = 0, 1, \dots, (a, m) - 1,$$

最大公因数个解.

1. $15x \equiv 9 \pmod{25}$

$$(15, 25) = 5 \nmid 9, \text{ 故无解.}$$

2. $3x \equiv 2 \pmod{7}$

$$(3, 7) = 1 \mid 2, \text{ 故只有 1 解.}$$

$$3x \equiv 1 \pmod{7} \quad 3x-1 \sim 7$$

解得 $x_0 \equiv 5 \pmod{7}$

$$3x \equiv 2 \pmod{7}$$

$$\Rightarrow x = 2x_0 = 10 \pmod{7} \\ = 3 \pmod{7}$$

3. $6x \equiv 3 \pmod{9}$

1. $(6, 9) = 3 \mid 3$, 有解.

$$2x \equiv 1 \pmod{3} \leftarrow \text{同除以 3 化简}$$

2. $x_0 \equiv 2 \pmod{3}$

3. $6x \equiv 3 \pmod{9}$

4. $x = 2 + 3t$

$$t = 0, 1, 2$$

$$\frac{3}{(6, 9)} \times x_0 \quad x \equiv 2, 5, 8 \pmod{9} \leftarrow \text{记得还原}$$

模重复平方法

$$b^n \pmod{m}$$

1. 将 n 写为 2 进制

$$n = n_0 \cdot 2^0 + n_1 \cdot 2^1 + \dots + n_n \cdot 2^n$$

$$1 \quad 2 \quad 4 \quad 8 \quad 16 \quad 32 \quad 64 \quad 128 \quad 256 \dots$$

$$35 = 32 + 2 + 1 = 2^0 + 2^1 + 2^5$$

2. m, b , 令 $a = 1$

$$\text{若 } n_0 = 1 \quad a_0 = a \cdot b \quad b_0 = b^2 \pmod{m}$$

$$n_1 = 1 \quad a_1 = a_0 \cdot b_0 \quad b_1 = b_0^2 \pmod{m}$$

$$n_2 = 0 \quad a_2 = a_1 \quad b_2 = b_1^2 \pmod{m}$$

$$n_3 = 0 \quad a_3 = a_2 \quad b_3 = b_2^2 \pmod{m}$$

$\vdots$

$$n_n = 1 \quad a_n = a_{n-1} \cdot b_{n-1} \pmod{m}$$

$$\Rightarrow b^n \pmod{m} \equiv a_n \pmod{m}$$

$$312^{13} \pmod{667}$$

$$13 = 8 + 4 + 1 = 2^0 + 2^2 + 2^3$$

$$n_0 = 1 \quad a_0 = 1 \times 312 = 312$$

$$b_0 = 312^2 \equiv 629 \pmod{667}$$

$$n_1 = 0 \quad a_1 = 312$$

$$b_1 = 629^2 \equiv 110 \pmod{667}$$

$$n_2 = 1 \quad a_2 = 312 \times 110 = 34320$$

$$b_2 = 110^2 \equiv 94 \pmod{667}$$

$$n_3 = 1 \quad a_3 = 34320 \times 94 = 3226080$$

$$\equiv 468 \pmod{667}$$

中国剩余定理

$$\begin{cases} x \equiv b_1 \pmod{m_1} \\ x \equiv b_2 \pmod{m_2} \\ \vdots \\ x \equiv b_k \pmod{m_k} \end{cases}$$

$m_1, m_2, \dots, m_k$ 两两互素

$b_1, b_2, \dots, b_k$ 任意整数

$$x \equiv b \pmod{20} \Rightarrow \begin{cases} x \equiv b \pmod{4} \\ x \equiv b \pmod{5} \end{cases}$$

普通步骤

① 求 m

$$m = m_1 \cdot m_2 \cdot m_3 \cdots m_k$$

② 求 M_i

$$M_i = m / m_i$$

③ 求 M_i' (计算量巨大)

$$M_i' \cdot M_i \equiv 1 \pmod{m_i}$$

④ 代入表示解

$$x = b_1 M_1' M_i + b_2 M_2' M_2 + \cdots + b_k M_k' M_k \pmod{m}$$

计算方法 tips

消除同余式前的系数

$$ax \equiv b \pmod{m}$$

$$\Rightarrow ax \equiv b + my \pmod{m}$$

$$b + my = a \times 1 \quad \text{当 } y \text{ 有整数解时}$$

$$b + my = a \times 2 \quad \text{对应的倍数}$$

$$b + my = a \times 3$$

$$\Rightarrow x \equiv ? \pmod{m}$$

$$11x^2 \equiv -6 \pmod{7}$$

$$11x \equiv -6 + 7y$$

$$x \equiv 2 \pmod{7}$$

$$x^2 \equiv 2 \pmod{7}$$

$$23x \equiv 1 \pmod{140}$$

$$140 = 4 \times 5 \times 7$$

$$\Rightarrow \begin{cases} 23x \equiv 1 \pmod{4} & 3 \times 23 = 1 + 4y \\ 23x \equiv 1 \pmod{5} & 2 \times 23 = 1 + 5y \\ 23x \equiv 1 \pmod{7} & 4 \times 23 = 1 + 7y \end{cases}$$

$$\Rightarrow \begin{cases} x \equiv 3 \pmod{4} \\ x \equiv 2 \pmod{5} \\ x \equiv 4 \pmod{7} \end{cases}$$

求 M_i

$$m = 4 \times 5 \times 7 = 140$$

$$m = m_i M_i$$

$$\Rightarrow \begin{cases} M_1 = 35 \\ M_2 = 28 \\ M_3 = 20 \end{cases}$$

$$\begin{aligned} x &= b_1 M_1 M_1' + \cdots + b_k M_k M_k' \\ &= 907 \pmod{140} \\ &= 67 \pmod{67} \end{aligned}$$

求 M_i'

$$\Rightarrow \begin{cases} 35 M_1' \equiv 1 \pmod{4} \\ 28 M_2' \equiv 1 \pmod{5} \\ 20 M_3' \equiv 1 \pmod{7} \end{cases} \begin{cases} 35x - 1 = 4y & x=3 \\ 28x - 1 = 5y & x=2 \\ 20x - 1 = 7y & x=6 \end{cases} \Rightarrow \begin{cases} M_1' = 3 \\ M_2' = 2 \\ M_3' = 6 \end{cases}$$

中国剩余定理

算法优化 (欧拉定理及模重复平方法)

高次同余式的解数.

$$f(x) \equiv 0 \pmod{m} \quad \left\{ \begin{array}{l} f(x) \equiv 0 \pmod{m_1} \\ \vdots \\ f(x) \equiv 0 \pmod{m_k} \end{array} \right. \quad \begin{array}{l} \text{解:} \\ x_1 = 1, 2 \\ \vdots \\ x_2 = 2, 3, 4 \end{array} \quad \begin{array}{l} f(x) \text{ 的解数} = \\ \text{每个同余式解的} \\ \text{个数乘积} \end{array}$$

$$\begin{array}{cc} x_1 = 2, 3 & x_2 = 2, 3, 4 \\ \downarrow & \downarrow \\ x = b_1 \underline{m_1 m_1'} + b_2 \underline{m_2 m_2'} \Rightarrow 2 \times 3 = 6 \text{ 个解.} \end{array}$$

欧拉函数.

求解小于 m 的, 与 m 互素的个数.

1. $m = p^\alpha$

$$\varphi(m) = p^\alpha - p^{\alpha-1}$$

2. $m = a \times b, (a, b \text{ 互素})$

$$\varphi(m) = \varphi(a) \cdot \varphi(b)$$

3. 若 m 为素数

$$\varphi(m) = m - 1$$

$$\begin{aligned} \varphi(20) &= \varphi(4) \cdot \varphi(5) \\ &= (2^2 - 2) \times 4 \\ &= 8 \end{aligned}$$

$$\varphi(5) = 4.$$

$$\varphi(25) = 5^2 - 5 = 20$$

素数模的同余式简化、求解

$$f(x) = ax^n + \dots + a'x^n \equiv 0 \pmod{p}$$

$$r(0) = 0 \equiv 0 \pmod{p} \checkmark$$

$$\text{改写为 } f(x) = (x^p - x)q(x) + r(x) \Rightarrow$$

$$r(p-1) = a \equiv b \pmod{p} \times$$

取 $\equiv 0 \pmod{p}$ 的 x , 即为解

此时 $f(x)$ 与 $r(x)$ 等价

化简同余式

凑原式

通过后续处理将多余项消除

令 $g(x) = (x^7 - x)$ 町次数

$$2x^4 + 4x^3 + 2x^1 + x^1 + x^6 + x^3 + 12x^2 + x \equiv 0 \pmod{7}$$

$$(x^7 - x)(3x^7 + 4x^6 + 2x^4 + x^2 + 3x + 4) + 2x^5 + 2x^3 + 15x^2$$

$$-x^7 \quad -3x^3 \quad -4x^7 \quad -2x^5 \quad -x^3 \quad +3x^4 \quad -4x$$

凑到最高项后
括号外加凑

① 括号内 $(-x)$ 项

② 括号外加凑项

① + ② = 原项

$$f(x) = 2x^4 + 4x^3 + 2x^1 + x^1 + x^6 + x^3 + 12x^2 + x \equiv 0 \pmod{5}$$

$$(x^5 - x)(3x^9 + 4x^8 + 2x^6 + 3x^5 + 5x^4 + 2x^2 + 4x + 5) + 3x^3 + 16x^2 + 6x$$

$$-x^7 \quad -4x^9 \quad -2x^7 \quad -3x^6 \quad -5x^5 \quad -2x^3 \quad -4x^2 \quad -5x$$

$$r(x) = 3x^3 + 16x^2 + 6x \pmod{5}$$

$$r(1) = 25 \equiv 0 \pmod{5}$$

$$r(2) = 100 \equiv 0 \pmod{5}$$

$$r(3) = 279 \equiv 4 \pmod{5}$$

$\vdots$

故 $x = 1, 2 \pmod{5}$

欧拉判别

$$x^2 \equiv a \pmod{p}$$

$$a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$$

$$a^{\frac{p-1}{2}} \equiv -1 \pmod{p}$$

判断二次同余式的解. $\nearrow$

1. p 为奇素数 $x^2 \equiv a \pmod{p}$

勒让德符号

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{平方剩余} & \text{有解} \\ -1 & \text{平方非剩余} & \text{无解} \\ 0 & p \mid a \end{cases}$$

简便运算.

周期 $\left(\frac{a+p}{p}\right) = \left(\frac{a}{p}\right)$

可乘 $\left(\frac{a \cdot b}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$

若 $(a, p) = 1$, 则 $\left(\frac{a^2}{p}\right) = 1$

各情况如何算

$$\left(\frac{1}{p}\right) = 1$$

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} \quad \times$$

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$$

p, q 互素

$$\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}} \left(\frac{p}{q}\right)$$

$\uparrow$
二次互反律

2. 若 m 为非素奇数. (弱化条件, 扩大范围) $x^2 \equiv a \pmod{m}$ (奇数)

雅可比符号

$\downarrow$
① 先判断 $\left(\frac{a}{m}\right)$ $\begin{cases} \text{若为 } -1, \text{ 无解} \\ \text{若为 } 1, \text{ 不确定, 继续算} \end{cases}$

② 分解 $\Rightarrow \begin{cases} x^2 \equiv a \pmod{m_1} \\ x^2 \equiv a \pmod{m_2} \end{cases}$

$$\left(\frac{a}{m_1}\right) = 1 \text{ 且 } \left(\frac{a}{m_2}\right) = 1$$

同时有解.
才有解.

3. 简单计算二次同余式

$$x^2 \equiv a \pmod{X} \xrightarrow{X=x_1 x_2 x_3} \begin{cases} x^2 \equiv a \pmod{x_1} \\ x^2 \equiv a \pmod{x_2} \\ x^2 \equiv a \pmod{x_3} \end{cases}$$

是否有解.

1. $x^2 \equiv 7 \pmod{227}$

$$\frac{7}{227} = (-1)^{\frac{6}{2} \times \frac{226}{2}} \cdot \frac{227}{7}$$

$$-1 \cdot \frac{3}{7} = (-1) \cdot \frac{1}{3} = 1$$

解数

有

2. $x^2 \equiv 11 \pmod{511}$

$$\begin{aligned} \frac{11}{511} &= (-1)^{\frac{10}{2} \cdot \frac{510}{2}} \cdot \frac{511}{11} \\ &= (-1) \cdot \frac{5}{11} \\ &= (-1)(-1)^{\frac{4}{2} \cdot \frac{10}{2}} \cdot \frac{1}{5} \\ &= (-1) \cdot \frac{1}{5} \\ &= -1 \end{aligned}$$

无

11. $x^2 \equiv -6 \pmod{91}$

$$x^2 \equiv 16 \pmod{91}$$

$$\frac{16}{91} = \frac{4^2}{91} = 1$$

相关, 有解.

$x^2 \equiv 2 \pmod{67}$

$$\frac{2}{67} = (-1)^{\frac{67^2-1}{8}} = -1$$

无解

$x^2 \equiv 2 \pmod{37}$

$$\frac{2}{37} = (-1)^{\frac{37^2-1}{8}} = -1$$

无解.

$x^2 \equiv -2 \pmod{67}$

$$\begin{aligned} (-1) \times \frac{2}{67} &= (-1)^{\frac{67^2-1}{8}} \cdot (-1) \\ &= 1 \end{aligned}$$

$x^2 \equiv -2 \pmod{37}$

$$\begin{aligned} \frac{-1}{37} \times \frac{2}{37} &= (-1)^{\frac{37^2-1}{2}} \times (-1) \\ &= -1 \end{aligned}$$

无解?

求模的原根

原根: $a^e \equiv 1 \pmod{m}$

最小整数 $\rightarrow e = \text{ord}_m(a)$

$$a^{\varphi(m)} \equiv 1 \pmod{m}$$

a 为模 m 的原根

$$a^e \equiv 1 \pmod{7}$$

$\downarrow \quad \downarrow$

底数 次数

$\varphi(m)$ 个

与 m 互素的数

1 2 3 4 5 6

$\varphi(m)$ 的因数

6 的因数

1 2 3 6

$\Rightarrow$

a 1 2 ③ 4 ⑤ 6

$\text{ord}_m(a)$ 1 3 6 3 6 2

$e = \varphi(7) = 6$ 原根

个数 $= \varphi(\varphi(7)) = 2$ 个

① $b \equiv a \pmod{m}$

$$\text{ord}_m(b) = \text{ord}_m(a)$$

② $a^{-1} \cdot a \equiv 1 \pmod{m}$

$$\text{ord}_m(a^{-1}) = \text{ord}_m(a)$$

求模的所有原根 $\varphi(\varphi(m))$ 个

1. 找出 1 个原根 a

2. $a^{x_1} \pmod{m}$

$$a^{x_2} \pmod{m}$$

$$a^{x_3} \pmod{m}$$

$$a^{x_4} \pmod{m}$$

x 是与 $\varphi(m)$ 互素的数

求模 17 的所有原根

1. $\varphi(17) = 16$